

Week 5 Assessment

Object Detection, Object Tracking & Sports Video Analytics

Project-Based Assessment | Total Marks: 100

Assessment Question — Sports Match Analyzer

Develop a complete sports video analytics system using a recorded match video. Choose **ANY ONE**: **Option A — Tennis Match Analyzer** OR **Option B — Badminton (Shuttle) Match Analyzer**.

Core Technologies

- Python and OpenCV
- YOLO / Ultralytics for object detection
- An object tracking method covered during the course
- Recorded sports match video
- Court ROI and zone analysis
- Annotated video generation and saving

Common Processing Flow

```
Recorded Match Video
↓
YOLO Object Detection
↓
Player Detection
↓
Object Tracking + Persistent IDs
↓
Court ROI / Court Zones
↓
Position History
↓
Movement & Video Analytics
↓
Annotated Output Video + Match Summary
```

Option A — Tennis Match Analyzer

Develop a video analytics application for a recorded tennis match.

Minimum Requirements

- Detect tennis players using YOLO / Ultralytics.
- Track players across frames using persistent IDs.
- Define the tennis court ROI.
- Determine and store player positions.
- Draw player trajectories.
- Divide the court into meaningful zones.
- Generate movement and court analytics.
- Save the final annotated output video.
- Tennis ball detection and tracking may be implemented when a suitable model/video is available.

Option B — Badminton (Shuttle) Match Analyzer

Develop a video analytics application for a recorded badminton match.

Minimum Requirements

- Detect badminton players using YOLO / Ultralytics.
- Track players across frames using persistent IDs.
- Define the badminton court ROI.
- Determine and store player positions.
- Draw player trajectories.
- Divide the court into logical front, middle and back zones.
- Generate movement and court analytics.
- Save the final annotated output video.
- Shuttle detection and tracking may be implemented when feasible with the selected model/video.

Mandatory Video Analytics Outputs

Implement all of the following for either selected project:

No.	Analytics Output	Expected Result
1	Player Detection & Persistent Tracking ID	Detect players and maintain IDs across frames.
2	Player Position	Store/display player position on the court.
3	Player Trajectory	Visualize the movement path across frames.
4	Total Movement Distance	Calculate cumulative movement from position history.
5	Movement Speed	Estimate instantaneous and/or average speed.
6	Court Zone Detection	Identify the current zone of each player.
7	Zone Occupancy	Calculate time or percentage spent in each zone.
8	Most Active Zone	Identify the most frequently occupied zone.
9	Movement Heatmap / Position Density	Visualize frequently visited court locations.
10	Annotated Output Video	Display detections, IDs and selected analytics and save the video.

Advanced Analytics — Choose Any THREE

Implement at least **three** additional analytics:

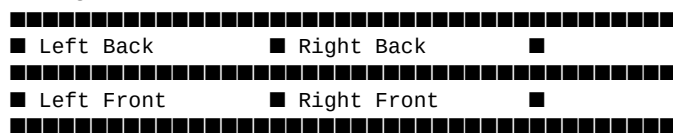
1. Average speed
2. Maximum speed
3. Court coverage percentage
4. Unique zones visited
5. Zone entry / exit count
6. Zone transition analysis
7. Front vs back court occupancy
8. Left vs right court occupancy
9. Activity time vs stationary/inactive time
10. Player-to-player distance
11. Average/minimum/maximum inter-player distance
12. Player movement comparison
13. Court coverage comparison
14. Player heatmap comparison
15. Ball/shuttle position and trajectory
16. Ball/shuttle speed
17. Player-to-ball/shuttle distance
18. Nearest player to ball/shuttle
19. Rally/activity detection
20. Rally duration
21. Longest/average rally
22. High-activity periods
23. Low-activity or break periods
24. Movement direction statistics

Suggested Visual Outputs

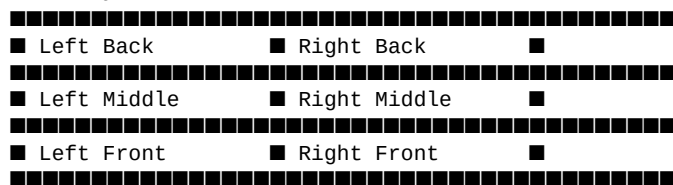
- Bounding boxes with class names and tracking IDs
- Player center point and position coordinates
- Trajectory trails
- Court ROI and zone overlay
- Current speed and total distance overlay
- Current zone indicator
- Movement heatmap / position density map
- Optional ball or shuttle trajectory
- Final match analytics summary

Suggested Court Zone Layouts

TENNIS



BADMINTON



Submission Requirements

- Complete source code / Jupyter Notebook.
- Recorded input video or a valid public dataset/resource reference.
- Annotated output video saved successfully.
- At least one visual analytics output such as heatmap, trajectory visualization or zone analysis.
- Brief algorithm flow and explanation of the implemented analytics.
- Final summary of the selected analytics results.

Assessment Evaluation

Component	Marks
YOLO Detection and Object Tracking	20
Court ROI, Zones and Position Analysis	15
Mandatory Video Analytics	30
Three Advanced Analytics	15
Visualization and Annotated Output Video	10
Code Quality, Explanation and Final Summary	10
Total	100

Final Instruction: Choose any one project — Tennis Match Analyzer or Badminton Match Analyzer. Implement all mandatory analytics and at least three advanced analytics.